Poster Abstract: A Configurable Piezoelectric Acoustic Metasurface for Real-Time Tracking

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***Abstract*—Acoustic metasurfaces have emerged as a promising approach for enhancing acoustic communication and sensing, yet existing designs remain impractical in mobile scenarios due to two key limitations. First, 3D-printed metasurfaces are inherently static, while reconfigurable designs rely on slow, power-hungry mechanical actuators, hindering real-time operation. Second, prior systems often assume known target directions or rely on device feedback, which fails in non-cooperative, device-free settings. In this paper, we present a configurable piezoelectric metasurface that enables low-power, real-time acoustic signal control, along with a passive direction-estimation method for localizing non-cooperative users. Experimental results show microsecond-level reconfiguration, and simulations demonstrate a median direction estimation error below 1°.**

***Index Terms*—Piezoelectrical Metasurface, Real-time Tracking**

I. Introduction

Recent years have witnessed rapid progress in acoustic metasurfaces for enhancing both acoustic communication and sensing [1]–[3]. By engineering the phase or amplitude response of incident sound waves through an array of subwavelength meta-atoms, an acoustic metasurface can reshape propagation without actively generating signals. Leveraging this capability, prior work has shown improvements in communication link [1], sensing range [2], and imaging quality [3].

Despite these advances, existing acoustic metasurfaces are largely designed for static deployments and offer limited adaptability in mobile environments. Two factors drive this gap. First, 3D-printed metasurfaces are fixed after fabrication, while reconfigurable designs typically rely on slow mechanical actuation [2], [4], making them both latency- and energy-prohibitive for real-time mobility. Second, most prior systems assume either prior knowledge of the target direction or target-side feedback, assumptions that rarely hold in mobile sensing, where targets are often non-cooperative and device-free.

To address these challenges, we present a digitally programmable acoustic metasurface built on piezoelectric resonators, eliminating bulky and power-hungry mechanics and enabling microsecond-level reconfiguration. We further exploit this fast programmability to estimate the direction of a passive target. Specifically, we cycle through a small set of predefined metasurface codebooks (i.e., per–meta-atom phase configurations) and measure the resulting reflections at the transceiver. This produces a codebook–direction–signal mapping that allows us to infer the target's direction without cooperation or feedback. With microsecond reconfiguration, we can track direction changes and steer beams in real time, making metasurface-assisted mobile sensing practical.

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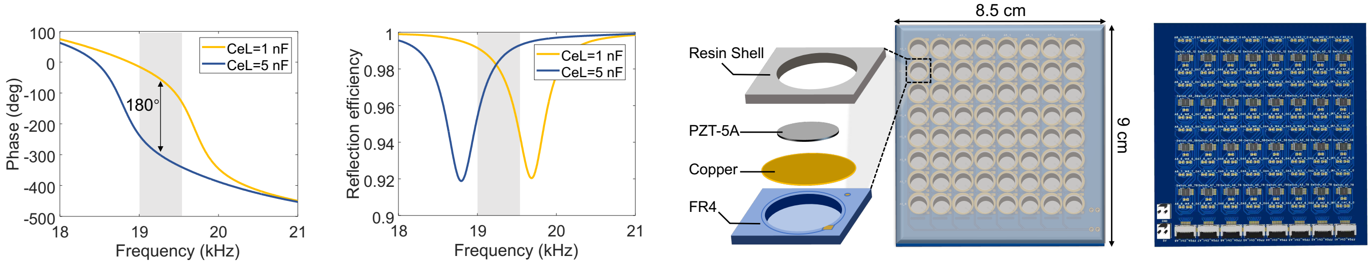
II. System Design

To enable metasurfaces to operate in mobile settings, we design a piezoelectric metasurface with microsecond-level programmability and develop a beam-tracking algorithm that continuously adapts to moving targets.

Meta-atom Design. We build each phase-shifting meta-atom using piezoelectric materials. Due to intrinsic electromechanical coupling, the acoustic reflection depends on both the device structure and the external electrical load impedance (e.g., connected capacitors). By tuning the load impedance, we directly control the reflection phase for real-time reconfiguration. Figure 1(c) shows the meta-atom design: a copper-backed piezoelectric element mounted inside a resonant cavity. The cavity acoustically couples to the piezoelectric resonator, allowing the meta-atom's effective impedance (reflection phase) to be controlled in the acoustic domain. By optimizing the cavity geometry, the meta-atom achieves a larger phase tuning range within our operating band. As shown in Figure 1(a), varying the capacitor changes the load impedance, $Z_{ce} = \frac{1}{i\omega C_{eL}}$, thereby shifting the reflection phase across different frequencies. Our results show up to a 180° phase shift with high reflection efficiency ($> 90\%$) within the operating band.

Metasurface Assemble and Control. The metasurface is built by assembling many optimized meta-atoms. Figure 1(c) and (d) show the detailed meta-atom structure and an 8×8 prototype, respectively, where each meta-atom is paired with an electronic switch that selects between two load capacitors, enabling two discrete phase states (0 and π). A custom driver board on the backside (Figure 1(e)) integrates 32 dual-channel switch ICs (TMUX1134PWR) and 128 capacitors. Each meta-atom is controlled by one GPIO pin of an MCU, resulting in 64 GPIOs for fully parallel control. The GPIO lines directly provide bias voltages, enabling microsecond-level reconfiguration due to fast switching and small RC time constants.



(a) Phase shift. **(b) Reflection efficiency.** **(c) Meta-atom** **(d) Top View** **(e) Bottom View**

Fig. 1: Design of the metasurface. (a) and (b) show the phase shift and reflection efficiency of the meta-atom, respectively. (c)-(e) show the meta-atom, the assembled metasurface, and the bottom control board, respectively.

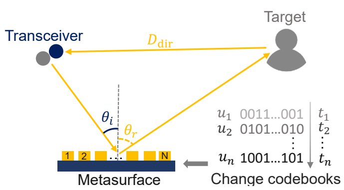


Fig. 2: Channel model for metasurface-based direction estimation.

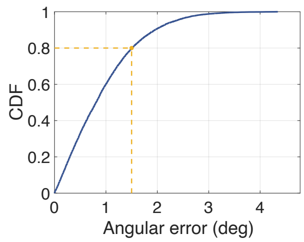


Fig. 3: Simulated angular error.

Localization of an Unknown Target. To localize the target, we program the metasurface with a sequence of codebooks to construct a measurement matrix $\mathbf{U} = \{u_1, u_2, \cdots, u_P\}$. This produces multiple measurements that retain target angle-dependent information. Figure 2 shows the channel model for metasurface-based direction estimation. We consider a metasurface with N meta-atoms and assume both the transceiver and target are in the far field. Under the p-th codebook, the p-th measurement is:

$$\mathbf{h_p} = \sigma \sum_{n=1}^{N} e^{j\varphi_n^p} \cdot e^{j\varphi_n^{in}(\theta_i)} \cdot e^{j\varphi_n^{ref}(\theta_r)} + \mathbf{h_{dir}} + \mathcal{N}, \quad (1)$$

where σ denotes the combined attenuation, including path loss along the "transceiver–metasurface–target–transceiver" path and the reflection amplitude of the n-th meta-atom. φ_n^p is the phase shift of the n-th meta-atom under the p-th codebook. $\varphi_n^{in}(\theta_i)$ and $\varphi_n^{ref}(\theta_r)$ denote the incident and reflected phases at the n-th meta-atom, respectively. Here, θ_i can be measured in advance, and θ_r is the target direction to be estimated. h_{dir} denotes the direct target–transceiver channel, and $\mathcal{N}$ is additive white Gaussian noise.

Since $\mathbf{h}_{\text{dir}}$ is invariant across codebooks, it can be treated as a constant interference term. Subtracting the sample mean over all $P+1$ measurements, the received signal vector $\mathbf{H}$ is given by:

$$\mathbf{H} = \mathbf{U}\mathbf{A}(\theta) + \boldsymbol{\mathcal{N}}$$
$$= \sigma e^{j\varphi_n^{in}(\theta_i)} \begin{bmatrix} e^{j\phi_1^1} & e^{j\phi_2^1} & \cdots & e^{j\phi_N^1} \\ e^{j\phi_1^2} & e^{j\phi_2^2} & \cdots & e^{j\phi_N^2} \\ \vdots & \vdots & \vdots & \vdots \\ e^{j\phi_1^P} & e^{j\phi_2^P} & \cdots & e^{j\phi_N^P} \end{bmatrix} \begin{bmatrix} 1 \\ e^{jkd\sin\theta} \\ \vdots \\ e^{jk(N-1)d\sin\theta} \end{bmatrix} + \boldsymbol{\mathcal{N}}. \quad (2)$$

Based on Equation 2, we optimize the codebooks to minimize mutual coherence:

$$\rho(\mathbf{u}_i, \mathbf{u}_j) = \frac{\mathrm{cov}(\mathbf{u}_i, \mathbf{u}j)}{\sigma_{\mathbf{u}i}\ \sigma_{\mathbf{u}_j}} \leq \delta, \quad (3)$$

where $\mathbf{u}_i$ and $\mathbf{u}_j$ are two arbitrary codewords, and $\delta = 0.2$ is the threshold. This yields a measurement matrix $\mathbf{U}$ to recover the high-resolution direction-of-arrival estimation via atomic norm minimization. Leveraging microsecond-scale reconfiguration, we can sweep these codebooks fast enough to estimate the target direction in real time. Once θ is obtained, we reconfigure the metasurface to beamform toward the target.

We conduct simulations to evaluate the localization performance at various angles. The simulation results are shown in Figure 3. We can see that the median angular error is less than $1°$, and the 80th percentile error is about $1.5°$, demonstrating the feasibility of the localization algorithm.

III. Future Work.

In this section, we highlight two key challenges for future work. *1). How to design codebooks that maximize directional discrimination and signal SNR?* To enable long-range localization, the codebook should achieve both high gain and low mutual coherence (i.e., good RIP). Existing methods often rely on prior knowledge of the target direction, which is impractical, motivating designs that work without directional priors. *2). How to separate target reflections from environmental interference?* In practice, metasurface reflections can be overwhelmed by multipath, static reflectors, and interference, degrading direction estimation. A key challenge is to reliably isolate the metasurface–target component from these dominant signals.

References

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